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Acute Abdominal Pain

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Abdominal pain is common and often minor. Severe abdominal pain that comes on quickly, however, almost always indicates a significant problem. The pain may be the only sign of the need for surgery and must be attended to swiftly. Older adults and people who have HIV infection or who are taking immunosuppressants (including corticosteroids) may have less abdominal pain than younger or healthy adults with a similar disorder, and, even if the condition is serious, the pain may develop more gradually. Young children, especially newborns and infants, may develop abdominal pain but are unable to communicate the reason for their distress.

Types of Abdominal Pain

There are different types of abdominal pain depending on the structures involved.

Visceral pain comes from the organs within the abdominal cavity (which are called the viscera). The viscera's nerves do not respond to cutting, tearing, or inflammation. Instead, the nerves respond to the organ being stretched (as when the intestine is expanded by gas) or surrounding muscles contract. Visceral pain is typically vague, dull, and nauseating. It may be hard to pinpoint. Upper abdominal pain results from disorders in organs such as the stomach, duodenum, liver, and pancreas.

Midabdominal pain (near the navel) results from disorders of structures such as the small intestine, upper part of the colon, and appendix. Lower abdominal pain results from disorders of the lower part of the colon and organs in the genitourinary tract.

Somatic pain comes from the membrane (peritoneum) that lines the abdominal cavity (peritoneal cavity). Unlike nerves in the visceral organs, nerves in the peritoneum respond to cutting and irritation (such as from blood, infection, chemicals, or inflammation). Somatic pain is sharp and fairly easy to pinpoint.

Causes of Acute Abdominal Pain

Pain can arise from many causes, including infection, inflammation, ulcers, perforation or rupture of organs, muscle contractions that are uncoordinated or blocked by an obstruction, and blockage of blood flow to organs.

Immediately life-threatening disorders, which require rapid diagnosis and surgery, include

- Ruptured [abdominal aortic aneurysm](#)
- [Perforated stomach or intestine](#)
- Blockage of blood flow to the intestine ([mesenteric ischemia](#))
- Ruptured [ectopic pregnancy](#)

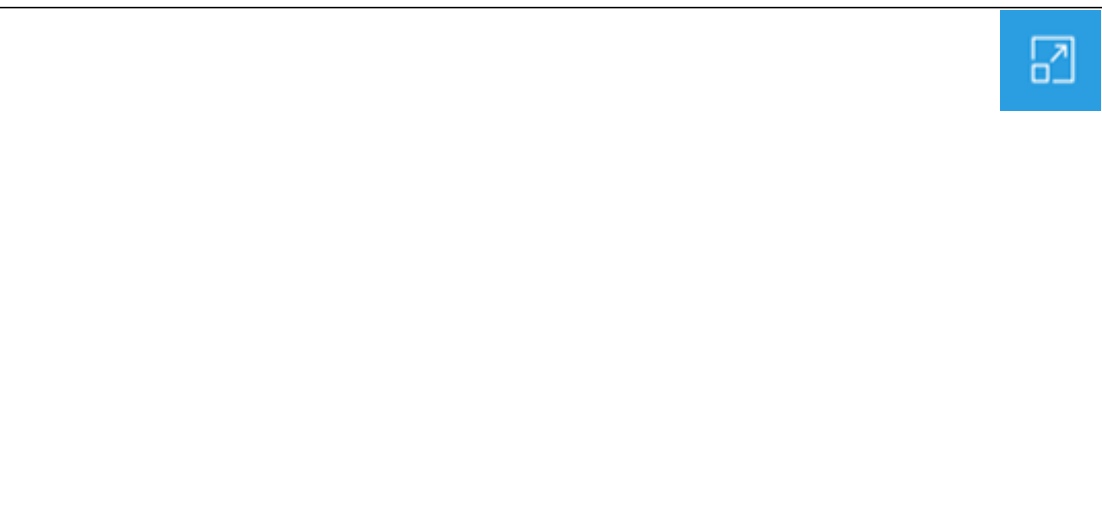
Serious disorders that are nearly as urgent include

- [Intestinal obstruction](#)
- [Appendicitis](#)
- Sudden (acute) inflammation of the pancreas ([pancreatitis](#))

Sometimes, disorders outside the abdomen cause abdominal pain. Examples include heart attack, pneumonia, and twisting of a testis ([testicular torsion](#)). Less common

problems outside the abdomen that cause abdominal pain include diabetic ketoacidosis, [porphyria](#), [sickle cell disease](#), and certain bites and poisons (such as a [black widow spider bite](#), heavy metal or methanol poisoning, and some [scorpion stings](#)).

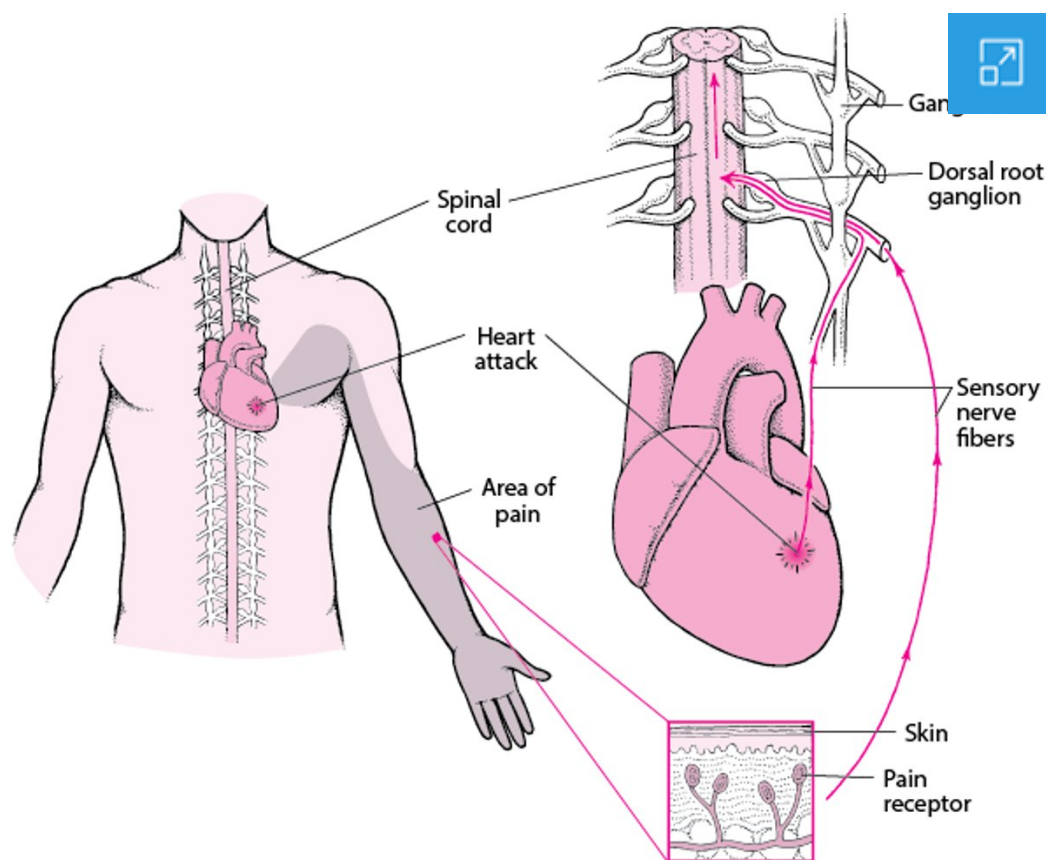
Abdominal pain in newborns, infants, and young children has numerous causes not encountered in adults.



Referred pain is pain that is felt in an area that is distant from the source (see figure [What Is Referred Pain?](#)). For example, a person who has gallbladder disease may feel pain in the shoulder blade. The source of the pain is the gallbladder, which is located in the abdomen, but the pain is felt in the shoulder.

What Is Referred Pain?

Pain felt in one area of the body does not always represent where the problem is because the pain may be referred there from another area. For example, pain produced by a heart attack may feel as if it is coming from the arm because sensory information from the heart and the arm converge on the same nerve pathways in the spinal cord.



Peritonitis

Peritonitis is inflammation of the peritoneal cavity.

It is very painful and almost always signals a very serious or life-threatening disorder. It can result from any abdominal problem in which the organs are inflamed or infected. Common examples include [appendicitis](#), [diverticulitis](#), and [pancreatitis](#). Also, blood and body fluids (such as intestinal contents or urine) are very irritating when

they leak into the peritoneal cavity and can cause peritonitis. Disorders that cause blood and body fluids to leak include spontaneous organ rupture (such as a [perforated intestine](#) or ruptured [ectopic pregnancy](#)) and severe abdominal injury.

People who have fluid in the peritoneal cavity (the fluid is called [ascites](#)) are at risk of developing an infection. Such an infection is called spontaneous bacterial peritonitis.

Once peritonitis has been present for a number of hours, the inflammation causes fluid to leak into the abdominal cavity. The person may then develop [dehydration](#) and go into [shock](#). Inflammatory substances released into the bloodstream may affect various organs, causing severe lung inflammation, kidney failure, liver failure, and other problems. Without successful treatment, people may die.

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Some Causes of Abdominal Pain in Newborns, Infants, and Young Children

Cause of Pain	Description	Comments
Meconium peritonitis	Inflammation and sometimes infection of the abdominal cavity and its lining (peritonitis) caused by a perforation in the intestine and leakage of meconium (the dark green fecal material that is produced in the intestines before birth)	Occurs while infants are still in the womb or shortly after birth
Hypertrophic pyloric stenosis	A blockage at the stomach outlet (duodenum)	Forceful (projectile) vomiting occurs after feedings Usually begins between birth and 4 months of age
Volvulus	Twisting of a loop of the intestines	Causes intestinal obstruction and cuts off the blood supply to intestines Commonly, vomiting, diarrhea, abdominal swelling, and episodic and excessive crying (colic)
Anorectal malformation (imperforate anus)	Narrowing or blockage of the anal opening	Normally detected by doctors when infants are examined after birth and usually requires immediate surgery



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